Solving quadratic equations by factoring

1 Solve the following equations by using the zero product property:

a $x(x+6)=0$

b $-3m(m+2)=0$

c $x(7x-12)=0$

d $(x-7)(x-6)=0$

e $(x-7)(x+8)=0$

f $(17x-19)(11x-5)=0$

g $(7x+19)(2x-5)=0$

h $(4x-17)(3x+11)=0$

i $(4x-9)^2=0$

j $(3-3x)^2=0$

k $\frac{m}{6}(m-10)=0$

2 Solve the following equations by first factoring the left hand side of the equation.

a $x^2+12x=0$

b $3y-15y^2=0$

c $10x^2+40x=0$

d $x^2+19x+90=0$

e $x^2+6x-55=0$

f $x^2-17x+72=0$

g $x^2-20x+100=0$

h $x^2-5x-14=0$

i $4x^2+8x-32=0$

j $-x^2+x+20=0$

k $5x^2+22x+8=0$

l $-4x^2+25x-36=0$

m $4x^2-17x+15=0$

n $4x^2+17x+15=0$

o $5x^2+14x+8=0$

p $3x^2-7x-20=0$

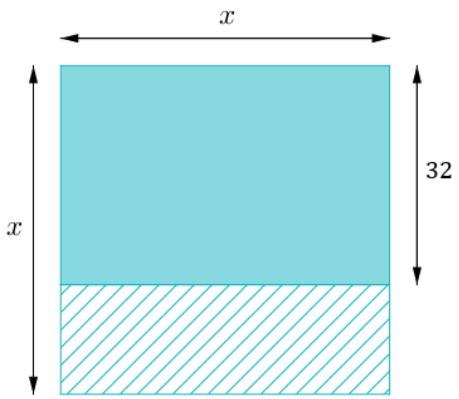
q $x^2-64=0$

r $81x^2-64=0$

3 Homer needs a sheet of paper x cm by 32 cm for an origami koala. The local origami supply store only sells square sheets of paper.
The lower portion of the image below shows the excess area A of paper that will be left after Homer cuts out the x cm by 32 cm piece. The excess area, in cm^2, is given by the equation $A=x(x-32)$.

a At what lengths x will the excess area be zero?

b For what value of x will Homer be able to make an origami koala with the least amount of excess paper?



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4 Solve the following equations by first rearranging, and then factoring:

a $m^2 - 27m = -182$

b $m^2 = m + 20$

c $m^2 = 14m$

d $11m^2 = 7m$

e $(y+1)^2 = 4y + 4$

f $\dfrac{2x^2 - 19x}{3} = 20$

5 Find the value of a:

a The nonzero solution of the equation $x(x+a) = 0$ is $x = 10$

b The nonzero solution of the equation$x(x+a) = 0$ is $x = -1$.

6 The equation $x(x-7) = -10$ has a positive integer solution $x = 5$. Determine the other solution.

7 Solve for the x-values of the x-intercepts of the parabola whose equation is$y = x^2 + x - 6$.

8 Software engineers are designing a self-serve checkout system for a supermarket. They notice that the traffic through the store during the day is described by the function $C = -t(t-12)$, where C is the number of customers and t is the number of hours after the store opens.

a To meet the peak demand, the engineers allow for an extra checkout machine to automatically turn on when the number of customers first reaches 32 people, and to automatically turn off when it next falls below 32 people.
Find the times t when the number of customers is equal to 32 people.

b Hence, the extra checkout machine will turn on ▯ hours after opening, and it will then turn off ▯ hours later.

9 The equation $4x^2 + kx + 16 = 0$ has one solution $x = 2$. Determine the value of the coefficient k.

10 Consider the quadratic expression $ax^2 - 32x - 80$. If you substitute $x = 4$ into the expression, what must be the value of a to make the expression equal 0?
Begin with the equation $ax^2 - 32x - 80 = 0$.

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11 State whether the following is a solution to the quadratic equation.

$$0 = x^2 - 3x - 10$$

a 1 **b** 2 **c** 4 **d** 5

Additional questions

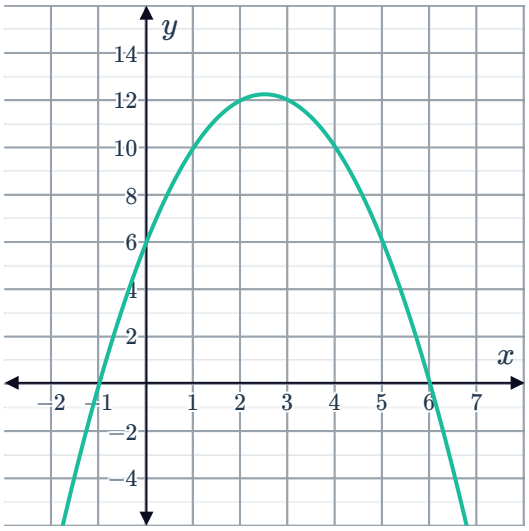
12 For each graph, complete the following:

i Determine the y-intercept.

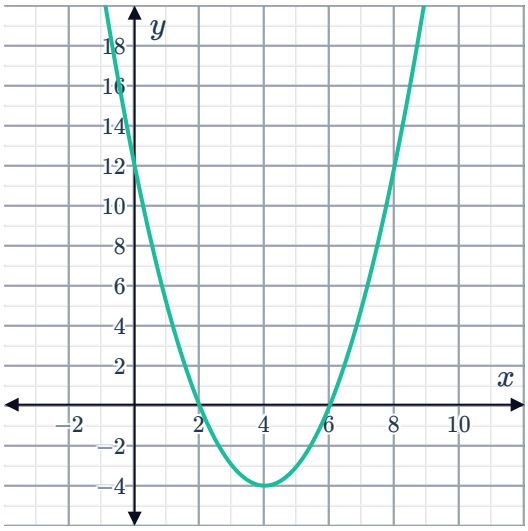
ii Determine the x-intercept(s).

iii Write the equation in factored form.

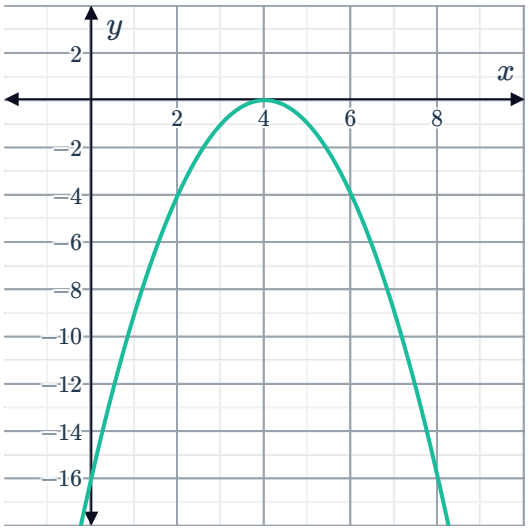
a



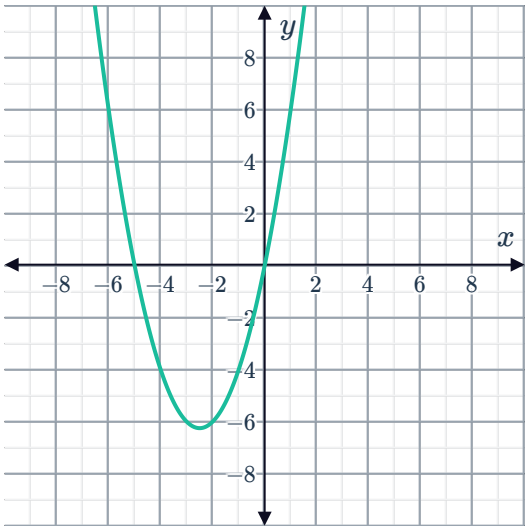
b



c

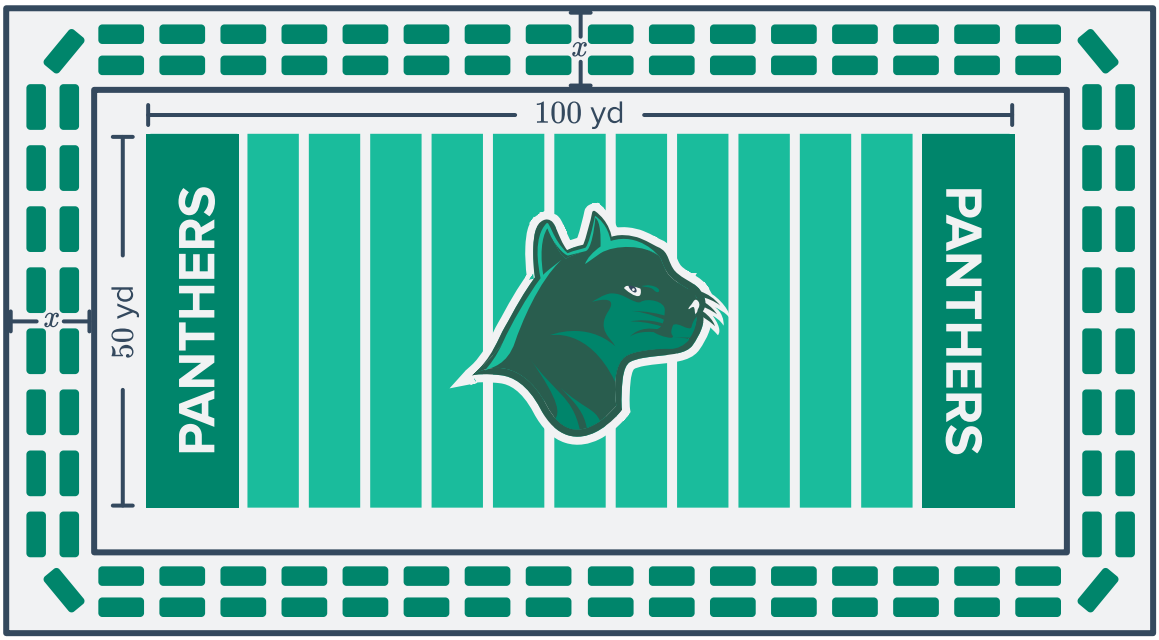


d



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13 The school football field is in the shape of a rectangle and has stadium seating all the way around the field that is of a uniform width. If the total area of the field and the stadium is $8400\ \text{yd}^2$, determine the width of the stadium seating.



$A = 8400\ \text{yd}^2$

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